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Generics in Java

=> Generics :-

➔ Generics means parametrized types which means that we can provide any type of parameter to the classes, interfaces or methods

➔ Generics were introduced in JDK 1.5 version

➔ Generics are represented by angular braces - < >

➔ The main objective of Generics are :-

1. To provide type safety
2. To resolve type casting problem

➔ By default "arrays are type safe"

➔ Now for collections, till JDK 1.4 version, collections were not generic types

➔ In JDK 1.5 version, Generic Collections were introduced

➔ NOTE : We can only provide non-primitive values in generics

=> Generic Classes :-

- ➔ If any class is declared with type parameters then it is known as Generic Class
- ➔ Generic classes can be user-defined classes or predefined classes (collections classes)

- ➔ Generic type can be any valid identifier name
- ➔ We can provide any number of parameters in generics

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=> Generic Bounded Types :-

- ➔ We can bound the type parameter for a particular range by using extends keyword. And

this concept is known as Generic Bounded Type Concept

➔ Syntax : class A<T extends X>
(X can be any class or interface)

➔ Cases :-

1. We can only use extends keyword, not implements keyword
2. We can only use Non-Primitive data types
3. class A<T extends X & Y>

Smoe time the on the function which type of data pass that time they are used in the ? mark

example

```
class A
{
  void show(ArrayList<?>al)
 {
   System.out.println(al)
 }
}

class Demo
{
  public static void main(String[] args)
  {
    A obj =  new A();

   ArrayList<Integer>al=new ArrayList<Integer>();
  al.add(23);

  ArrayList<String>al1 = new ArrayList<Integer>();
  al11.add("Vishal");
  }
}
```

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Company Name: Smart Programming (+91 62838-30308)

Address : Chandigarh & Mohali (Punjab), India

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